

PROJECT DESIGN PHASE-II

Technology Stack (Architecture & Stack)

Date:	29/09/26	Team ID:	SWTID-2026-8956
Project Name:	WeatherWise AI	Maximum Marks:	4 Marks
Team Size:	4 Members		
Team Members:	SANMATHI KUMAR (Leader), ARIFA ZOHRA, AINUL LIYANA P M, VAISHALI S		

1. TABLE-1: COMPONENTS & TECHNOLOGIES

S.No	Component	Description	Technology
1	User Interface / Client	HTTP client tools and web/mobile interfaces interfacing with system endpoints	Postman / Thunder Client
2	Application Logic (Runtime)	Asynchronous, event-driven server runtime environment	Node.js (v16+)
3	Web Framework	RESTful routing middleware and controller dispatching	Express.js
4	AI Recommendation Model	Natural language recommendation engine for weather-based advice	Google Gemini AI SDK (`@google/genai`)
5	Database ODM & Store	Document collection schemas and object modeling layer	MongoDB & Mongoose ODM
6	External API Services	Live temperature, humidity, and weather metric retrieval	OpenWeatherMap API
7	Authentication & Security	Stateless token authentication and route guarding	JSON Web Tokens (JWT)
8	Password Protection	Cryptographic hash salt algorithm for user credentials	`bcryptjs`
9	Infrastructure Requirements	Execution environment requirements	Windows 10/11, macOS, or Linux (8GB RAM, i5 CPU)

2. TABLE-2: APPLICATION CHARACTERISTICS

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Built using community-driven JavaScript runtime libraries.	Node.js, Express.js, Mongoose
2	Security Implementations	Route authentication middleware, bcrypt password encryption, parameter sanitization.	JWT, bcryptjs, Express Sanitization
3	Scalable Architecture	Decoupled 3-tier MVC architecture enabling easy microservices conversion.	Model-View-Controller (MVC) Pattern
4	Availability & Resiliency	Built-in Fallback Mode for basic execution without external API keys.	Custom Resilient Fallback Handlers
5	Performance Optimizations	Non-blocking I/O execution with compound database indices `{ user: 1, city: 1, country: 1 }`.	Async/Await, MongoDB Compound Indexing